

Oliver Ma

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Science in Electrical Engineering & Computer Sciences

Expected Graduation: May 2028

- **GPA:** 3.91/4.00

- **Relevant Coursework:** Data Structures; Signals, Dynamical Systems, and Information Processing; Data Science

Y Combinator

San Francisco, CA

Startup School 2026

June 2026 – June 2026

EXPERIENCE

Capital One

June 2026 – Present

Software Engineer Intern

McLean, VA

- Replace Capital One's static end-of-year email with an interactive "Year-In-Review" experience by developing Go Backend for Frontend (BFF) microservice to aggregate customer rewards data for millions of loyalty customers
- Reduce peak API latency by designing nightly ETL batch job that pre-computes reward recaps into PostgreSQL, resulting in constant read-time responses instead of expensive on-demand aggregation under high traffic
- Deliver zero-downtime production deployment by containerizing Go service with Docker and configuring CI/CD pipeline to AWS ECS Fargate with CloudFormation, resulting in automated build, test, and deploy workflows

Towards Data Science

Sep. 2024 – Oct. 2024

Contributing Writer

San Francisco, CA

- Authored technical article on YOLOv8 computer vision achieving 7,000+ views, providing practitioners with guidance on custom dataset training, model deployment, and implementation of object detection solutions
- Published content amplified across TDS social media channels, establishing thought leadership in computer vision and object detection methodologies, and driving significant readership, discussion, and community engagement

University of California, San Diego

July 2024 – Aug. 2024

Autonomous Vehicles Researcher

San Diego, CA

- Engineered scale autonomous vehicle achieving real-time self-navigation and object classification by integrating OpenCV computer vision libraries with NVIDIA Jetson embedded Linux platform and ROS2 operating system
- Deployed edge AI inference pipeline using Luxonis OAK-D depth camera and DepthAI APIs, processing 30+ FPS for obstacle detection, path planning, lane detection, and hand gesture recognition without cloud connectivity
- Designed and 3D-printed custom mechanical components including camera mounts and sensor housings

Verizon Communications

June 2023 – July 2023

Data Engineer Intern

San Jose, CA

- Architected and deployed full-stack DevOps monitoring dashboard integrating Jenkins and Jira APIs
- Developed real-time data visualization features enabling administrators to monitor Jenkins jobs and Jira issues/tickets at a glance, reducing administrator workflow time by consolidating metrics into single interface

PROJECTS

TraceFund

- Used Scaffold-ETH 2, Next.js, and wagmi to build full-stack Web3 dApp that eliminates donation fraud by locking funds in smart contract escrow until creators submit evidence and donors vote to release funds for each milestone
- Engineered custom smart contract using Solidity and deployed on Base Mainnet and various crypto wallets

OncoVision

- Engineered progressive web application using Svelte, Tailwind CSS, and TensorFlow achieving 85%+ accuracy in oral and skin cancer risk detection from camera image scans, securing Third Place at E-Med Hacks 2024
- Optimized CV model for real-time mobile inference, enabling instant health screening without cloud dependencies

PUBLICATIONS

The Comprehensive Guide to Training and Running YOLOv8 Models on Custom Datasets, Towards Data Science

02 Oct. 2024

TECHNICAL SKILLS

Languages/Frameworks: Java, Python, C/C++, SQL, Scheme, TypeScript, HTML/CSS, Node.js, React, Next.js